

Executive Data Analysis Report

Project Identifier: OLIST-LATE-DELIVERY-RISK

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1. Executive Summary

The model predicts instances of late delivery in orders with an OVERALL ACCURACY of 74.3%, correctly classifying 74.3% of all orders. The model separates cases of late delivery from on-time orders with an ROC-AUC of 0.769. At the standard 0.5 threshold, the model correctly identified 1,021 orders as late delivery but failed to catch 544 orders where late delivery actually occurred, and falsely flagged 4,416 on-time orders as late delivery. In total, the model captured 1,021 out of 1,565 cases of late delivery present across the evaluated orders.

2. Data Diagnostics Profile

Input shape	96,476 rows × 23 columns
Output shape	96,476 rows × 22 columns
Duplicate rows removed	0
High-missing columns flagged	None
Low-variance columns dropped	order_status
Columns type-corrected	None
Numeric columns imputed	customer_zip_code_prefix, n_items, n_distinct_sellers, n_distinct_products, total_price, total_freight, avg_product_weight_g, avg_product_photos_qty, total_payment_value, max_installments, n_payment_methods, purchase_to_estimated_days, is_late_delivery
Categorical columns imputed	order_id, order_status, order_purchase_timestamp, order_estimated_delivery_date, customer_unique_id, customer_city, customer_state, primary_seller_state, product_category, primary_payment_type

3. Model Performance Leaderboard

Task type: binary_classification | **Best model:** HistGradientBoostingClassifier

Candidate Model	CV Score
HistGradientBoostingClassifier	0.5721
RandomForestClassifier	0.5660
LogisticRegression	0.4749

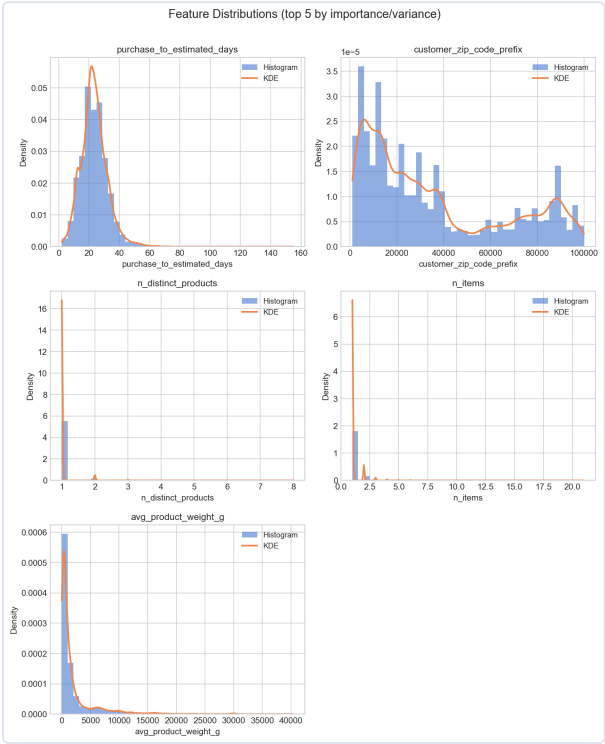
F1 (macro)	0.5673
Precision (macro)	0.5743
Recall (macro)	0.7017
ROC-AUC	0.7693

	Predicted Negative (0)	Predicted Positive (1)
Actual Negative (0)	13,315	4,416
Actual Positive (1)	544	1,021

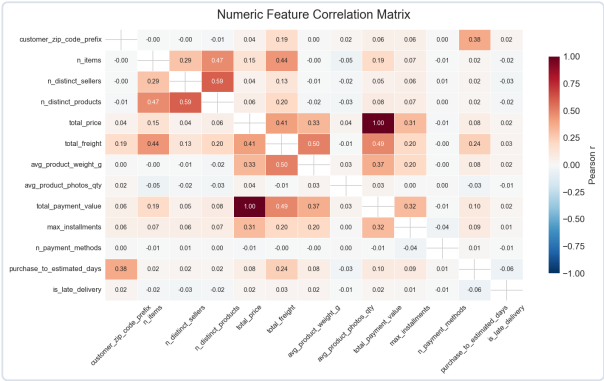
Threshold	Precision	Recall	F1 (macro)
0.3	0.114	0.897	0.379
0.4	0.149	0.802	0.494
0.5	0.188	0.652	0.567

4. Automated Business Insights

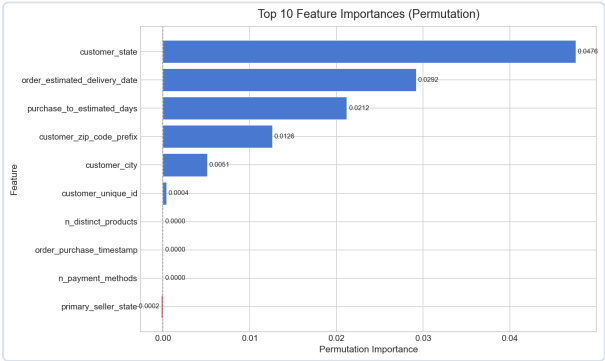
The `customer_state` matters most for predicting late delivery cases because regional logistics networks and local carrier coverage vary significantly, meaning late delivery depends heavily on the state where the order is delivered. The `order_estimated_delivery_date` is the second most relevant factor for cases of late delivery, as orders with shorter deadlines create logistical pressure that increases the risk of late delivery. The `purchase_to_estimated_days` is the third most relevant factor, since the number of days between purchasing an order and the expected delivery date influences the complexity of fulfillment, where longer transit windows can create scheduling bottlenecks that lead to late delivery. The `customer_zip_code_prefix` is the fourth most relevant factor for late delivery instances, because specific local routing areas have unique delivery challenges and carrier reliability differences that affect whether late delivery occurs for those orders.



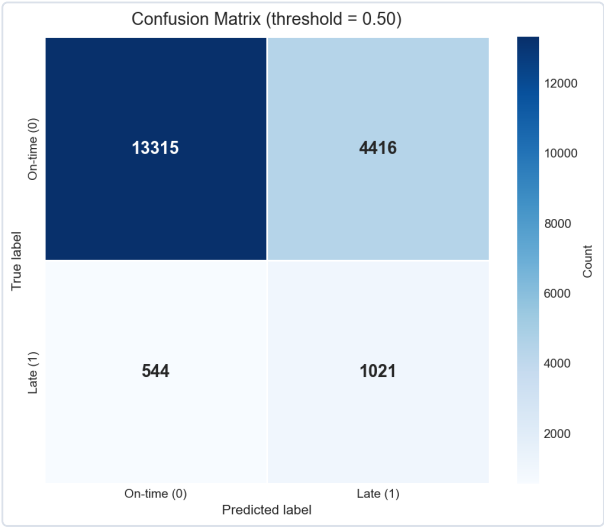
Distributions



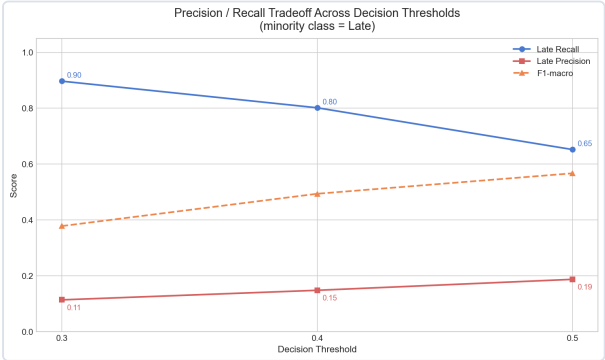
Correlation Heatmap



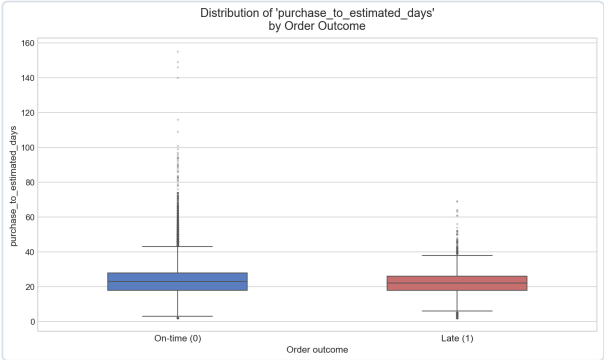
Feature Importance



Confusion Matrix



Threshold Tradeoff



Top Feature Vs Target

5. Strategic Recommendations

To maximize the capture of late delivery instances, set the threshold to 0.3, which yields a recall of 0.897, ensuring that 89.7% of orders are reviewed for late delivery, even though the precision of 0.114 means many flagged orders will not actually be late. To prioritize investigation efficiency and reduce false alarms for manual review teams, use the 0.5 threshold, which provides a precision of 0.188, meaning an 18.8% chance that a flagged order is actually a case of late delivery, while maintaining a recall of 0.652 that catches 65.2% of late delivery occurrences. To balance catching the majority of instances of late delivery with a reasonable screening rate, apply the 0.4 threshold, which achieves a recall of 0.802 to catch 80.2% of these orders, paired with a precision of 0.149.